The ECSU-PCE Dataset: A comprehensive recording of embodied social interaction with EEG, peripheral physiology, and behavioral measurements in adults

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Abstract

The study of dyadic social interaction has become an active field of research, yet many foundational questions regarding the mechanistic basis and potential role of inter-bodily and inter-brain synchrony remain. One methodological stumbling block has been the absence of a comprehensive, high-quality dataset in the public domain that could serve as a shared reference point for the field for the specific purpose of testing new hypotheses and methods. Here we provide such a dataset focused on embodied dyadic interaction, the ECSU-PCE Dataset, which includes sensorimotor variables and a range of physiological measurements: electroencephalogram (EEG), electrocardiogram (ECG), electrodermal activity (EDA), and respiration dynamics. Participants' subjective reports on their social experience and questionnaire-based personality traits are also included. The data was acquired from a total of 32 pairs of participants involving 64 healthy able-bodied adults. Each pair completed 18 one-minute trials of a haptics-based social contingency detection task, known as the "perceptual crossing experiment" (PCE); the current study successfully replicated previous PCE findings. Three-minute resting state data was also recorded at four points in time: before the start of social interaction, and after trials 6, 12, and 18, i.e. after the end of their interaction. The ECSU-PCE dataset is expected to be of interest for researchers interested social interaction, embodied cognition, and the effects of online technological mediation.

Introduction

Cognitive science has undergone an interactive turn in social cognition research. In particular, there is increasing interest in investigating the real-time brain-body dynamics underlying human social interaction. For this purpose, so-called "hyperscanning" approaches that permit simultaneous recording of the physiological activity of two or more participants have become popular. However, the guiding hypothesis of hyperscanning research, namely that interbrain synchrony is the mechanism that facilitates social interaction by enabling functional integration across multiple brains, has recently come under scrutiny. One central concern has been the lack of standardized tasks and methods in the field that could serve to more tightly constrain the testing of its hypotheses [1]. In this dataset report, we respond to this concern by releasing a comprehensive hyperscanning dataset focused on embodied dyadic interaction, the ECSU-PCE Dataset.

The "Perceptual Crossing Experiment" (PCE) is an experimental paradigm for the study of social cognition in the context of embodied social interaction, focusing on the minimal conditions necessary for mutual recognition in a haptically mediated virtual space. Originally developed by Lenay and colleagues [2], the PCE has already contributed to our understanding the embodied nature of social interaction, demonstrating that people can recognize the presence of another sentient agent through dynamic and interactive patterns alone [3]. Later studies have further explored the PCE's implications, including investigations into sensorimotor dynamics [4] and subjective experience of the other's presence [5–8] (Preprint Schizophrenia). The paradigm has been applied to different populations of participants, such as individuals with high-functioning autism [9], and adolescents [10]. To date, no studies have investigated the neural dynamics underlying task performance in the PCE paradigm. Yet the PCE's emphasis on a minimalist approach to the study of spontaneous interaction aligns well with the needs of hyperscanning approaches to social neuroscience for a standardized, highly controllable, yet also open-ended, and dynamically rich paradigm. The suitability of the PCE for this purpose is further underlined by the robustness of its results: this ECSU-PCE dataset replicates previous findings regarding the behavioral factors influencing participants' reported clarity of perceptual awareness of the other player.

In the hyperscanning PCE reported here, participants were situated in isolated rooms, preventing visual or auditory contact with each other. Interaction between partners was facilitated solely through the human-computer interface depicted in Fig 1; for a full technical report, see [11] (Preprint Technical Report). This interface allows each participant to move an avatar within a one-dimensional circular virtual space, effectively a line that connects end-to-end. It is important to note that the other objects located in this one-dimensional circular virtual space are not visible. Each participant can encounter two distracting objects: (i) a motionless static object positioned randomly, and (ii) a dynamic object that "shadows" the movement of the partner's avatar at a constant offset (see Fig 1). The interface's motor generates a binary vibrational signal: it is vibrating whenever there is an overlap with an object in the virtual environment, otherwise it remains off. However, the vibration is the same regardless of the object being crossed.

The task for each participant is to identify the moment when they recognize they are in interaction with their partner's avatar and to mark this moment for the experimenters with a button press (hereafter referred as "click"). A click is not perceivable by the other participant. Crucially, the only way for participants to distinguish between the objects in the virtual circle is through the distinctive patterns of interaction that they afford. When crossing the static and shadow objects, there is a unidirectional interaction and hence the vibration is only felt by the participant that is encountering those objects. In contrast, when participants are crossing each other's avatar, the perceptual interaction becomes bidirectional, and both receive the haptic feedback simultaneously.

Material and Methods

The Perceptual Crossing Device

The Perceptual Crossing Device comprises a console and two interfaces with haptic controllers Fig 2; for full details, see [11] (Preprint Technical Report). Supplementary files for the PCE device can be found in the following repository: (https://osf.io/9ecxu/files/osfstorage)

Participants

The total sample included 64 participants divided into 32 pairs. Participants from 24 nationalities, who are able to understand and communicate about the experiment in English, were included. The ages ranged from 18 to 68 years old (mean: 28.6 ± 8.9), and 39 of the participants identified as female. Efforts were made to ensure that participants were kept separate before the beginning of the experiment so that they did not meet with whom they would be interacting with. There are three exceptions: pair number 32 included two classmates, and pairs number 15 and number 25 were couples. The experiment was approved by the OIST Human Subject Research Review Committee (HSR-2023-013). All participants gave their informed consent to participate in the study.

Recruitment and exclusion criteria

Participants were recruited via public advertisements, including posters and flyers on the OIST campus, as well as online advertisements like internal mailing lists, social networks, and the research group's website.

Eligibility for the study was determined through a pre-experiment screening form that included questions about age (older than 18 years old), psychiatric or neurological conditions, intake of psychoactive substances (including medication), and allergic reaction to alcohol-based products (relevant for the EEG preparation).

Following the screening form, participants were required to provide personal information, select their preferred dates for participation, and complete the registration

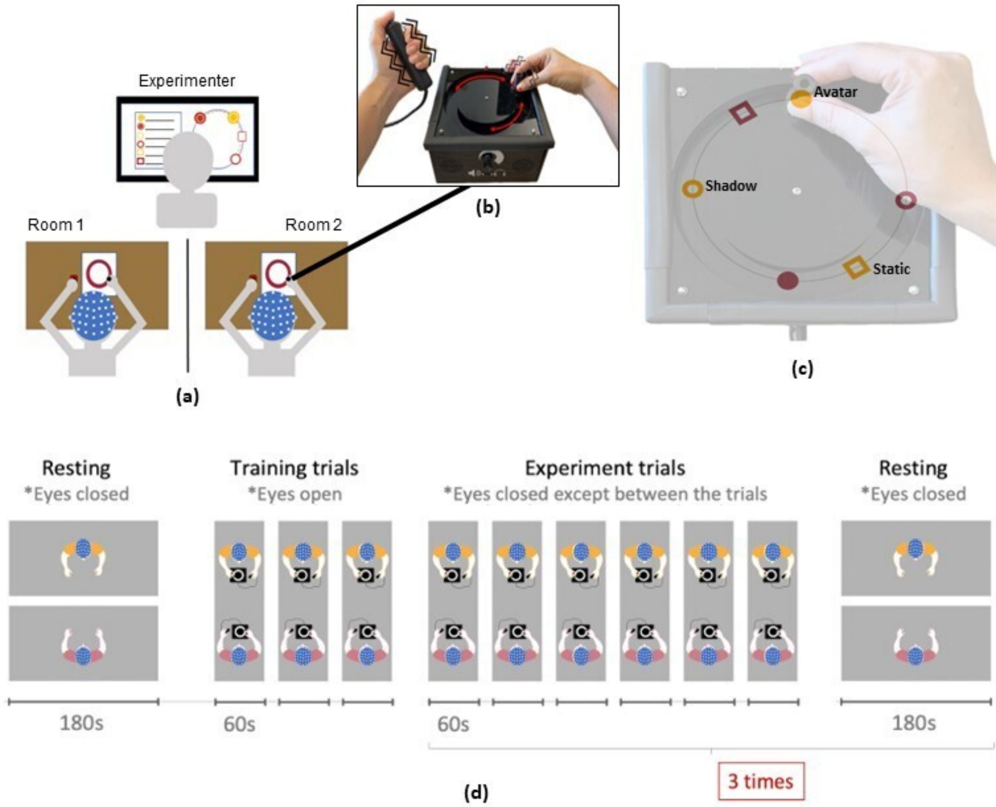

Fig 1. The experimental setup, device interface, and the experiment flow. (a) The PCE setup was arranged to ensure the participants were seated in a separate room to avoid contact with each other, apart from via each player's interface of the PCE device. (b) They were instructed to navigate the PCE device presented in front of them, namely to move the handle in rotational movement with one hand and to hold the button with the other hand, with no specific instruction regarding the use of dominant or non-dominant hand. (c) The location of the handle in the physical circle indicates the avatar's location in the virtually shared circle. (d) The experiment started with the initial resting state and was followed by 3 training trials and a total of 18 experimental trials, with a resting state performed after every 6 trials.

process. Prior to participation, participants were asked to avoid the use of any hair conditioner and hair products (foams, gels, sprays, etc.) the day before the scheduled appointment. They were also asked to avoid alcohol and caffeine consumption for the last 24 hours before and on the day of the experiment, respectively.

Fig 2. Display of the PCE device used to facilitate user interaction

Experimental protocol

Data collection happened from August to December 2023. Participants completed a single assessment that lasted approximately 3 hours and that consisted of two phases: preparation and recording. Once each of the participants arrived on site and both were seated in a separate room, the procedure of EEG preparation was performed on each participant (after the necessary documents and consent form were signed), starting about 1 hour prior to the beginning of the actual recording.

Preparation

When measuring brain activity, EEG preparation started with the head circumference measurement to ensure the right fit of the EEG cap. After the right size was found, an EEG cap along with 64 active electrodes (following the 10-20 standardised EEG electrode placement [12]) installed on it was placed on the participant's head. Once the EEG cap was securely fastened, the EEG conductive gel was applied to each electrode using a plastic syringe until the desired electrode impedance level was achieved.

Physiological sensor preparation, such as ECG and EDA electrodes, and the respiration belt, followed after all the EEG electrodes had reached the desired impedance level. Three ECG electrodes were prepared for 3-lead ECG recording. Prior to ECG electrodes placement, the participant was asked to clean their skin around both sides of the clavicles and lower edge of the left-side rib cage, by using a sterile all-purpose sponge and alcohol. Once the skin was cleaned, the electrodes were placed around those areas following the 3-lead ECG recording as follows: a positive electrode was placed below the right clavicle closer to the right shoulder, a ground electrode was placed on top of the left clavicle closer to the left shoulder, and a negative electrode was placed in between the ribs on the lower half of the left-side rib cage. After the ECG electrodes were securely attached, the respiration belt was placed around the participant's upper region of the abdomen. Finally, two EDA electrodes were placed on the skin of the participant's tips of two fingers (ring and index fingers) of whichever hand they had chosen to operate the button of their PCE interface.

Experimental protocol

The experimental protocol started with a resting state of 3 minutes to record brain activity through EEG. For this stage, participants were instructed to close their eyes and remain in a neutral sitting position. Afterwards, participants performed three training trials: the first two with eyes open and the last one with eyes closed. After the training phase a batch of 6 trials (1 minute each) was carried out, in which participants were instructed to close their eyes. After each trial participants were asked to answer different questions regarding their subjective experience. The questionnaires are described in detail below. After completing the first 6 trials, a second resting state period of 3 minutes took place. This resting state recording was repeated a third and fourth time after the completion of trials 12 and 18, so that in total 18 trials and 4 resting states were recorded. The experiment's sequence is depicted in Fig 1.

Psychological assessment

Big 5 personality traits evaluation

For this study, participants' personality traits were assessed, using Soto and John's Big Five Inventory-2 (BFI-2), before and after the experiment. The BFI-2 is designed to measure the five major dimensions of personality, known as the Big Five traits: Openness (this dimension reflects characteristics like imagination, creativity, and openness to new experiences), Conscientiousness (this includes traits such as organization, thoroughness, and a strong work ethic), Extraversion (this measures sociability, assertiveness, and emotional expressiveness), Agreeableness (this reflects attributes such as altruism, trust, and cooperativeness), and Neuroticism (this dimension assesses emotional stability and levels of anxiety, moodiness, or irritability). Each of these traits hence represents a range on a spectrum of attitudes, behaviors, and emotional patterns. The BFI-2 includes a set of statements that respondents rate according to how well they describe themselves, on a scale from "disagree strongly" to "agree strongly." This instrument has been shown to be a reliable and valid personality measure across different countries and is widely used in psychological research [13]. Participants were also asked to score their partner's personality based on their interaction during the experiment to see how closely the impression of their partner is to their partner's actual personality score.

Questionnaires about subjective experience

Following the completion of each trial, participants were presented with four different questions based on their in-trial actions. Question one, presented after every trial, inquired: "Did you experience your partner as missing, away, or absent at any moment during this trial? (No / Once / A few times / Many times / All the time)." The subsequent question, question two, contingent upon the response to the first, sought to gauge the clarity of the experienced absence if the initial answer was anything other than "No". This question asked: "On average, how clear did you experience that absence? (1 = No experience, 2 = Vague impression, 3 = Almost clear experience, 4 = Clear Experience)."

Further inquiries were made only if the participant clicked the button during the trial. Question three, employing the Perceptual Awareness Scale (PAS) — initially developed by [14] and later adapted for social perception studies by Froese et al. [5], and reused in 2020 [7] — queried: "How clearly did you experience that you were interacting with your teammate just before the time you pressed? Please select a category to describe the experience of your partner (1 = No experience, 2 = Vague impression, 3 = Almost clear experience, 4 = Clear Experience)". Lastly, question four, also derived from the work of Froese et al. (2014) [5], was posed as follows: "How confident are you now that you correctly identified a moment during which you were interacting with your partner? Please select a category to describe your confidence in your accuracy (1 = No confidence, 2 = Low confidence, 3 = Medium confidence, 4 = High confidence).

Questionnaires about the used strategy

After the completion of the 18 trials, additional items about the strategy used to solve the task were presented: "Describe any strategies you developed or used during the trials, with as much detail as possible. Describe any strategies you believe your partner developed or used, with as much detail as possible."

Brain activity and peripheral physiological data

EEG hyperscanning

The data was collected by using Brain Vision Brainamp DC amplifiers with PowerPack battery, standard 64 channels actiCAP snap set both by Brain Products (Brain Products GmbH, Gilching, Germany). The cutoff recording frequency was set from 0.016Hz to 1000Hz. Fpz was used as the ground electrode and FCz was used as the common reference electrode. The impedance was maintained below 25kΩ. The sampling frequency was set to 1000Hz and a notch filter was applied at 60Hz. EEG data was recorded by using BrainVision Recorder (Version 1.20.0502) software by Brain Products.

Continuous peripheral physiological recordings

ECG, EDA, and respiration dynamics (rate and amplitude) were collected by using Brain Vision BrainAmp ExG amplifier with PowerPack battery (Brain Products GmbH, Gilching, Germany). ECG signal was recorded in μV unit with a gradient of $0.1\frac{\text{mV}}{\mu\text{V}}$ and an offset of 0. EDA was recorded in microSiemens (μS) with a gradient of $25\frac{\text{mV}}{\mu\text{S}}$ and an offset of 0. Respiration rate was recorded in ARU (Arbitrary Respiration Unit) with a gradient of $1\frac{\text{mV}}{\text{ARU}}$ and an offset of 0. The sampling frequency was set to 1000Hz and a notch filter was applied at 60Hz to attenuate the power line frequency of 60Hz in Okinawa, Japan, where the data was collected. The peripheral physiological data were recorded by using BrainVision Recorder (Version 1.20.0502) software by Brain Products.

Software and code availability

All code that was implemented for data collection is available online at gitlab.com/oist-ecsu/pce-rpi, and continues to evolve. The version used for data collection was version 1.0: gitlab.com/oist-ecsu/pce-rpi v1.0.

Data Records

Data anonymization procedures

Upon registration on the website form, participants were assigned a randomly generated 4-digit generated ID. The table relating IDs and participants' names was only available to two members of administrative staff, so none of the experimenters had access to it. The dataset here presented does not contain any personal information nor the 4-digit IDs.

Behavioral data

The raw behavioral data contains 32 folders (one for each pair) named with the following convention: pceXXYYMMDD with XX being the pair number (starting from 01), and YYMMDD representing the date in which that particular pair was tested. For example, if a folder was named pce03230921 this would mean that it contains files from the third pair, and that the recording was made on the 21st of September of 2023. Within the folders there are three sub-folders: logs, trials, and questionnaires. The logs sub-folder contains information about the system specifications and they serve as control for bugs at different levels. The questionnaires sub-folder includes 4 files, 2 per participant: one containing the demographic information and the big 5 personality traits responses, and the other containing the responses to the questions about the strategy that were presented at the end of the 18 trials. The trials sub-folder contains 18 files which correspond to one of the each 18 trials that each pair carried out.

Table 1 presents the structure of a sample file from the trials sub-folder formatted as a Comma-Separated Values (CSV) document. This file contains several variables, including 'position', 'time', 'vibration', and 'button status'. Each observation in the file is labelled with an 'index', starting from 0, representing individual data points. The 'timestamp' variable captures the moment of data collection, with values ranging from 0 to 60 seconds for each experiment session. The file also records the positions of the static objects in the environment, identified as 'static object 0' and 'static object 1' for player 0 and player 1, respectively. The encoder translates the 1024 ticks to 600 unit, thus positions range from 0 to 599. The document tracks the motor vibrations through 'motor 0 vibrate software' and 'motor 1 vibrate software', using binary values (0 for no vibration and 1 for vibration). These vibrations occur when an avatar gets within 20 units of distance from another avatar, a shadow, or a static object. The current locations of the avatars are documented as 'pos0' and 'pos1'. The file also specifies the positions of the avatars' shadows as 'shadow delta0' and 'shadow delta1', where a shadow delta value of 150 means the shadow is positioned 150 units in front of the avatar. Lastly, 'button0' and 'button1' track whether the players have pressed their buttons, with 0 signifying no action and 1 indicating a button press.

Table 1. Perceptual crossing device collected data of a sample pilot trial.

index	timestamp	static_object_0	static_object_1	motor_0_vibrate_software	motor_1_vibrate_software	pos0	button0	shadow_delta0	pos1	button1	shadow_delta1
0	0	188	488	0	0	2.8125	0	150	416.7188	0	150
1	0.00135565	188	488	0	0	2.8125	0	150	416.7188	0	150
2	0.00254826	188	488	0	0	2.8125	0	150	416.25	0	150
3	0.00427876	188	488	0	0	3.28125	0	150	416.25	0	150
.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.
347	0.6966958	188	488	1	0	169.2188	0	150	395.625	0	150
348	0.7001844	188	488	1	0	169.6875	0	150	395.625	0	150
349	0.70394559	188	488	1	0	171.0938	0	150	395.625	0	150
350	0.7056012	188	488	1	0	172.0313	0	150	395.625	0	150
.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.
17979	36.6966231	188	488	1	1	105.4688	0	150	95.625	0	150
17980	36.6987406	188	488	1	1	104.5313	0	150	96.09375	0	150
17981	36.7008502	188	488	1	1	103.5938	0	150	97.03125	0	150
17982	36.7029609	188	488	1	1	102.6563	0	150	98.4375	0	150

EEG data and peripheral physiological data

Data structure

The peripheral physiological data and raw EEG data are organized in two folders. The first folder, 'Peripheral Physiological Data' folder contains 3 ZIP files namely 'ECG' contains ECG data, 'EDA' contains skin conductivity data, and 'RESP' contains respiration rate data. The peripheral physiological data provided in this article are all non-processed. Each of these ZIP files has 31 folders, each containing the data from one experiment. The second folder, 'Raw EEG Hyperscanning Data' folder contains raw/unprocessed EEG data and is organized into 31 ZIP files. Each zip file contains the data from one experiment. The preprocessed EEG hyperscanning data are accessible in Preprocessed EEG Hyperscanning Data Pt.1 (https://osf.io/bqs9h/) which contains 28 ZIP files from experiment 1 to 28, and in Preprocessed EEG Hyperscanning Data Pt. 2 (https://osf.io/p8n5u/) which contains 3 ZIP files from experiment 29, 30, and 32. Each zip file contains the data from one experiment. Each folder/ZIP file that contains data from one experiment has the data files of all tasks i.e. 4 resting states and 18 trials, from 2 participants, all in .mat format. Data from pair number 31 were not included in the published dataset due to incomplete recording caused by an error experienced by the data recording system. Data from pair 31 can be provided upon request when needed. A .ced file containing the EEG electrode arrangement used in the experiment is available to be downloaded in every folder containing EEG files.

EEG preprocessing

The EEG preprocessing was performed by using the MNE-python package (version 1.5) [15]. A zero-phase finite impulse response filter with an automatically set length was applied to the raw EEG data, i.e. a low pass filter at 60Hz, a high pass filter at 0.1Hz, and a notch filter at 60Hz. Visual inspection was later performed on the continuous data and any bad signals e.g. bad channels, noisy channels, broken/dead channels, were interpolated using spherical spline interpolation. Independent component analysis (ICA) with the number of components set at 30 was applied to the data that were high-pass filtered at 1Hz (ICA-specific filter) and the ICA solution was applied to the initial filtered signal (60Hz low pass and 0.1Hz high pass). All of the EEG signal preprocessing steps were done as per MNE guidelines (MNE guidelines for preprocessing). Visual inspection of ICA components to detect artefactual components was performed, which was guided by automatic detection of EOG using Fp1 and Fp2 and verified manually. After the removal of the artefactual components, the signal was reconstructed without the bad ICA components.

Recordings with special conditions

Five pairs of participants have incomplete data files due to various reasons. The conditions and reasons for these are as follows:

- Experiment pair pce01230807 only had the ECG and EDA data from Participant 1; due to a connection issue, the ECG and EDA data from Participant 2 were not properly recorded.
- Experiment pair pce09230828 had shorter Rest 1 (the first resting state) data (159,003 seconds instead of 180 seconds) in all EEG and peripheral physiological measurement for both participants, due to a late start in recording.
- Experiment pair pce24231107, only had ECG data from Participant 1. Due to a connection issue, the signal from Participant 2 was not properly recorded.
- Experiment pair pce26231121, had shorter Rest 1 (the first resting state) data (169,999 seconds instead of 180 seconds) in all EEG and peripheral physiological measurement for both participants, due to a "data buffer overflow error" briefly experienced by the system. The recording was resumed with the loss of around 10 seconds of continuous resting state data. This experiment also have ECG data

only from Participant 1; due to a connection issue, the signal from Participant 2 was not properly recorded.

- Experiment pair pce32230804, had ECG data only from Participant 2; due to a connection issue, the signal from Participant 1 was not properly recorded.

Variability in the quality of peripheral physiological signals may appear in some individuals. This is due to the nature of the experimental task that required movement, which may have caused the sensors' contact not to be optimal.

Behavioral analysis replication

The structure of the behavioral data recorded during the PCE is similar to that collected by Froese et al. (2020) [7], the main difference being that the current data included 32 dyads (rather than 10), and 18 trials (rather than 20) per dyad. Figure 3 graphically represents part of these data, showing the within-trial relationship between (a) the individuals' Perceptual Awareness Scale (PAS) clarity rating (see section Psychological assessment), (b) whether or not they recognized the other participant (as identified by correct or incorrect clicking, respectively) and (c) the time interval between the clicks of both participant's (inter-click interval); only trials where both members of the dyad produced a click are shown.

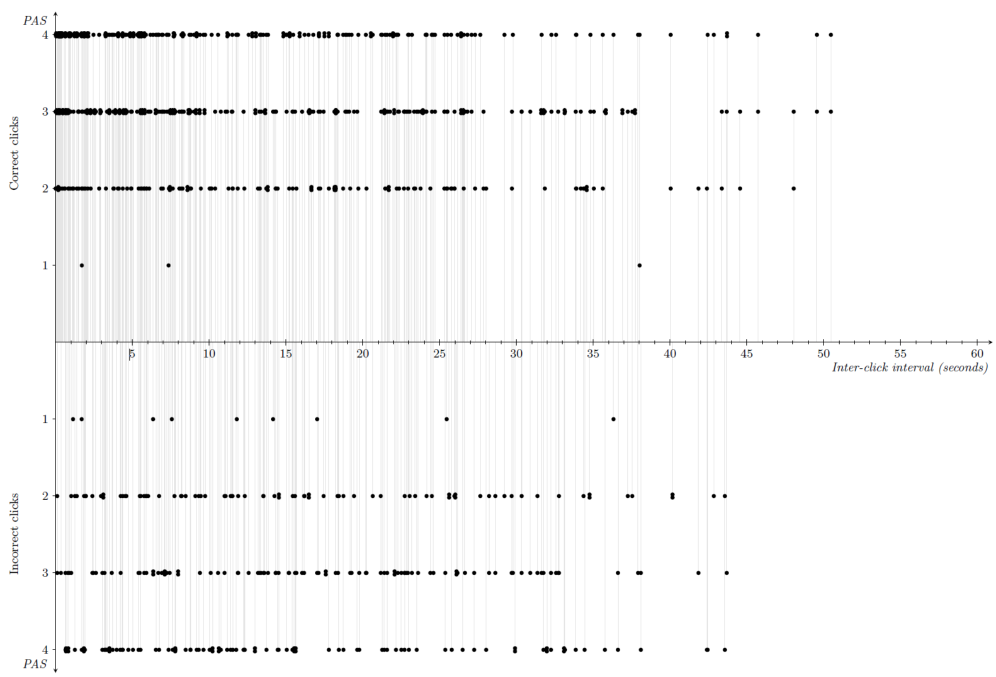

Fig 3. Graphical representation of the relationship between individuals' responses. The thin vertical gray lines connect both responses in a dyad that correspond to the same trial and projects the corresponding points to the horizontal axis to show the corresponding inter-click interval.

We fitted to these data the same statistical model as described in Froese et al. (2020) [7]. This model analyzes the relationships between the mentioned variables in the PCE. Specifically, the variables included in this model are the following (index i ($i = 1, \ldots, 32$) refers to the dyads, index j ($j = 1, \ldots, 18$) to the trials within a dyad, and index k ($k = 1, 2$) to refer to the players within a dyad):

- Recognizing the other player by making a correct click: We define X_{ijk} as a dichotomous variable that takes a value of 1 if player k of dyad i made a correct click (i.e., a click in response of crossing the avatar of the other player) during trial j, and 0 otherwise (i.e., clicking in response to a different event or not clicking at all).
- The rating given on the Perceptual Awareness Scale (PAS): Y_{ijk} is an ordinal variable with four categories that indicate the level of a player's awareness of the presence of the other player during the interaction (specifically, just before making a click). Here, we code Y_{ijk}, the PAS-response of player k of dyad i for trial j, in the following manner:

$$Y_{ijk} = \begin{cases} 0 & \text{if the player indicates “having had no experience of the other”,} \\ 1 & \text{if the player indicates “having had a vague impression of the other”,} \\ 2 & \text{if the player indicates “having had an almost clear experience of the other”,} \\ 3 & \text{if the player indicates “having had a clear experience of the other”.} \end{cases}$$

Note that Y_{ijk} is missing whenever the individual did not click during the trial.

Furthermore, at the level of the dyad, the following variables are defined:

- Short inter-click interval: The binary variable D_{ij} assumes a value of 1, if both players of dyad i clicked during trial j within a time frame of 3 seconds or less, and 0 otherwise (which includes the case that either of both players did not click). This definition applies irrespective of whether the players produced a correct click or not.
- Jointly recognizing the other player:

$$XX_{ij} = X_{ij1} \times X_{ij2}.$$

This analysis relates the participants' PAS responses in different trials with their clicking behavior and the time interval between the clicks of both participants in the trial, while also a learning process is modelled. Fig. 4 shows the relations between variables, the direction of the predictions made by the modelled and the statistical significance of the findings.

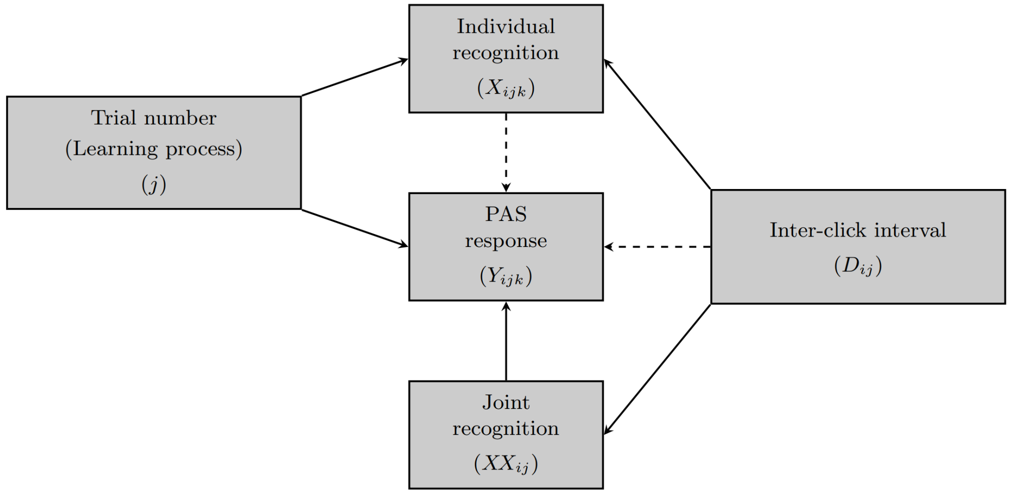

Fig 4. Relations among observed variables at the level of the dyad (joint recognition, inter-click interval), trial (trial number), and individual participant (individual recognition, PAS response). Dashed arrows indicate that for those relations no significant effect was found.

We will not repeat the description of the statistical model in this paper; all details, including the Bayesian procedure for estimation of the model's parameters, can be found in Froese et al.'s (2020) [7]. Table 2 shows the parameter estimates for this model adjusted to the data in Froese et al. (2020) as well as to the current data set. The main findings largely coincide across both data sets:

1. We hypothesized that, during the first trials, participants go through a learning process with respect to both individual recognition and their PAS response. The results in both studies confirm the hypothesis with respect to the learning process for the PAS response; with respect to the learning process for individual recognition, the current data set confirms the hypothesis, whereas for the 2020 study this was not the case.

2. The model included effects to predict the PAS response by individual recognition, joint recognition, and inter-click interval. In both studies, individual recognition and the inter-click interval do not show a significant effect, whereas joint recognition does reliably predict the PAS response. (One may note that the high-posterior density interval for parameter $\beta_{XX}^{(Y)}$, estimated from the current data, includes 0; however, the posterior probability that this parameter is larger than 0 turns out to be 97.5%.)

3. The hypothesis that the inter-click interval affects individual as well as joint recognition is reliably confirmed on both studies.

Table 2. Summary of the estimated marginal posterior distribution for the main parameters in the statistical model through the posterior mean and standard deviation, and the 95%-high posterior density interval.

	Froese et al. (2020)		**Current data**	
Parameter	Posterior Mean	95%-high posterior density interval	Posterior Mean	95%-high posterior density interval
${\beta^{(XX)}}_0$	-0.35	[-0.54; -0.16]	0.18	[0.05; 0.31]
$\beta_D^{(XX)}$	1.01	[0.40; 1.59]	0.86	[0.52; 1.20]
ρ_{XX}	0.52	[0.29; 0.74]	0.23	[0.08; 0.38]
${\mu^{(X)}}_0$	0.18	[-0.06; 0.43]	0.46	[0.32; 0.61]
$\sigma_0^{2(X)}$	0.15	[0.03; 0.33]	0.16	[0.06; 0.26]
${\mu^{(X)}}_1$	0.07	[-4.02; 4.71]	0.15	[0.00; 0.30]
$\sigma_1^{2(X)}$	29.35	[0.04; 88.50]	0.08	[0.01; 0.21]
$\mu^{(X)}$	0.12	[0.04; 0.21]	0.09	[0.00; 0.20]
$\nu^{(X)}$	4.91	[0.44; 13.78]	27.16	[0.88; 95.31]
$\alpha^{(X)}$	0.51	[0.07; 1.34]	2.24	[0.15; 8.62]
$\beta^{(X)}$	4.40	[0.26; 12.58]	24.92	[0.90; 88.47]
$\beta_D^{(X)}$	0.70	[0.16; 1.29]	0.89	[0.60; 1.19]
ρ_{YY}	0.04	[-0.22; 0.31]	0.16	[0.03; 0.30]
δ_{12}	1.53	[1.23; 1.82]	2.27	[1.97; 2.58]
δ_{23}	1.13	[0.92; 1.34]	1.50	[1.37; 1.64]
${\mu^{(Y)}}_0$	1.94	[1.36; 2.53]	-0.55	[-0.87; -0.23]
$\sigma_0^{2(Y)}$	0.57	[0.17; 1.12]	1.11	[0.68; 1.62]
${\mu^{(Y)}}_1$	0.63	[-0.01; 2.26]	0.30	[0.15; 0.46]
$\sigma_1^{2(Y)}$	0.58	[0.01; 1.06]	0.03	[0.01; 0.07]
$\mu^{(Y)}$	0.22	[0.00; 0.42]	0.14	[0.04; 0.26]
$\nu^{(Y)}$	19.73	[1.73; 50.93]	9.54	[1.39; 23.20]
$\alpha^{(Y)}$	4.03	[0.17; 12.41]	1.33	[0.22; 3.59]
$\beta^{(Y)}$	15.69	[0.94; 42.30]	8.21	[1.04; 20.00]
$\beta_X^{(Y)}$	-0.16	[-0.55; 0.20]	0.10	[-0.13; 0.34]
$\beta_{XX}^{(Y)}$	0.69	[0.31; 1.06]	0.21	[-0.01; 0.42]
$\beta_D^{(Y)}$	0.14	[-0.28; 0.54]	0.05	[-0.18; 0.28]

Appendix: Technical validation

Part of the value of the ECSU-PCE dataset lies in its high temporal resolution and low lag-times. In this section we report on the technical validation of these properties.

Sampling rate of behavioral data

The ECSU-PCE dataset comprises a total of $15,296,208$ data samples. It was observed that the average time increment between consecutive samples is approximately 1.77ms, as shown in Fig S1. The median time increment is approximately 1.66ms. The average sampling frequency amounts to 566.5Hz.

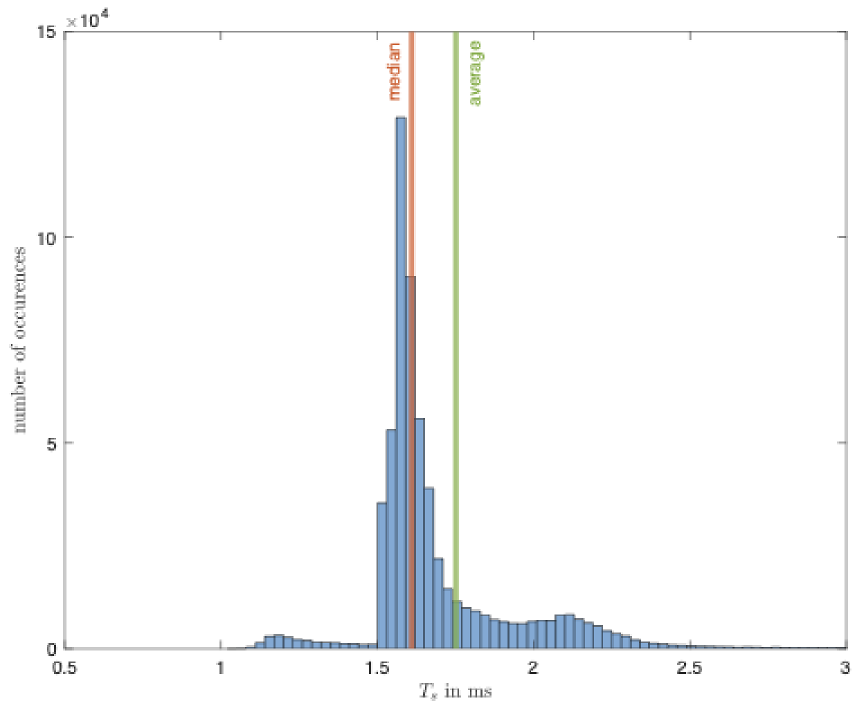

Fig S1. Sampling frequency of the data collected by the PCE device.

Encoder Latency

In order to move the avatars in the PCE virtual space, an encoder was utilised in the system controller. The encoder is capable of controlling the avatar's relative positioning instead of its spatial coordinates. The encoder has a resolution of 1024 pulse/revolution, meaning that because the software's virtual field is 600 units, a single pulse from the encoder equated to a movement of roughly 0.35° in the virtual space.

False positive and negative cases of motor vibration

It is important to address potential anomalies that may arise during data acquisition, such as instances where the signal to induce vibration is not sent when expected or, conversely, when it is erroneously transmitted. Detection of these false cases are essential to ensure the accuracy and reliability of the recorded data.

False negatives, wherein the signal to initiate vibration is expected but not observed, and false positives, where the signal is erroneously transmitted, are investigated separately. It was determined that the incidence of false negatives occurred in 7,358 samples, accounting for only 0.048 percent of the total dataset. This low occurrence rate underscores the robustness of the data acquisition system in reliably capturing instances when the vibration stimulus should be applied.

False positives were also examined, meaning that the signal to initiate vibration is observed although not expected. Our analysis revealed that the occurrence of false positives was observed in 58,669 samples, representing approximately 0.38 percent of the entire dataset.

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